

The Three Questions

Immanuel Kant, *Critique of Pure Reason* (1781)

Carlos Cotrini

What the five weekends behind us did

Weekend 0	NumPy, scikit-learn, PyTorch	the tools
Weekend 1	entropy, KL, variance, ensembles	how to measure what you do not know
Weekend 2	confounding, selection, attribution	why a model that scores well can be wrong
Weekend 3	attacks, jailbreaks, privacy	that somebody is trying to break it
Weekend 4	policy gradients, actor-critic, PPO	how a machine chooses under uncertainty

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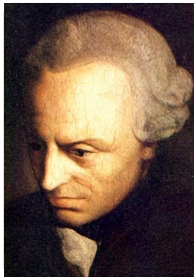
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Weekends 1 to 4 were about a model. Weekends 5 to 7 are about a system.

All the interest of my reason



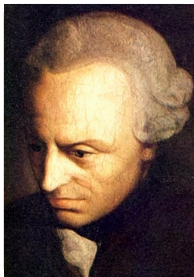
Immanuel Kant (c. 1790),
painter unknown

Alles Interesse meiner Vernunft (das spekulative sowohl, als das praktische)
vereinigt sich in folgenden drei Fragen:

- 1.
- 2.
- 3.

Kritik der reinen Vernunft, A805 / B833; translation C. Cotrini

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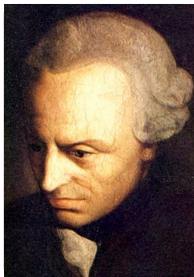
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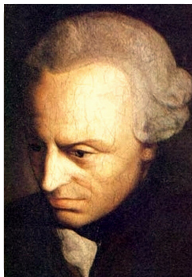
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What can I know? What ought I to do?

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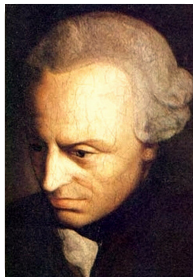
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Kant asked these of himself. We will ask them of the machine.

The same three questions, asked of the machine

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Was kann es wissen? → Weekend 5: RAG what it may know beyond its weights

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Was darf es hoffen? → Weekend 7: Large Scale AI what it may become, for what we spend

darf, not *wird*: not what will happen, but what it is **entitled to expect**, once we have done our part.

wenn ich nun tue, was ich soll, was darf ich alsdann hoffen?

A scaling law is a warrant for that expectation. It is not a theorem.